



Team Details

- Team name – **1 QNTX 3NOS**
- Team leader name – **VEERABABU JAKKULA**
- Problem Statement – **GENERATIVE AI FOR EVERYONE**

Brief about the idea

The idea is to build a free AI-powered Learning and Career Assistant using Generative AI that supports students throughout their learning and career journey. The platform helps users understand complex topics through simple explanations, provides summaries, quizzes, and practice questions, and creates personalized learning paths based on individual goals.

In addition to learning support, the system offers basic career guidance such as skill recommendations, resume improvement tips, and interview preparation. By removing cost and complexity barriers, the solution aims to make high-quality education and career assistance accessible to students from all backgrounds.

Opportunities

- How different is it from any of the other existing ideas?
- How will it be able to solve the problem?
- USP of the proposed solution

How is it different from existing ideas?

It combines learning, practice, and career guidance in one free, simple AI platform instead of multiple paid or complex tools.

How will it solve the problem?

By providing personalized explanations, quizzes, and guidance instantly using Generative AI, reducing confusion and learning time.

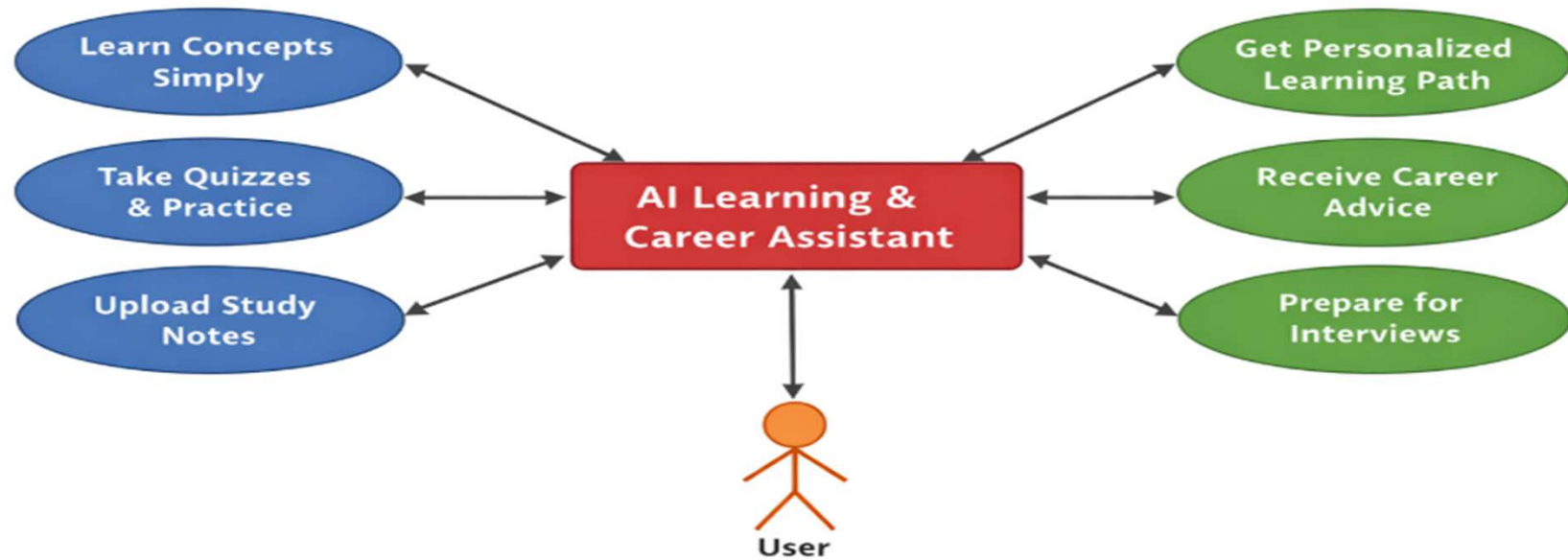
USP of the proposed solution

100% free, easy to use, personalized learning and career support in a single AI-powered platform.

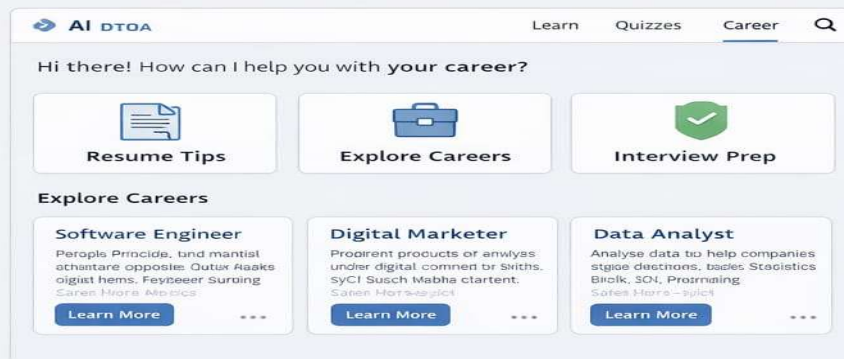
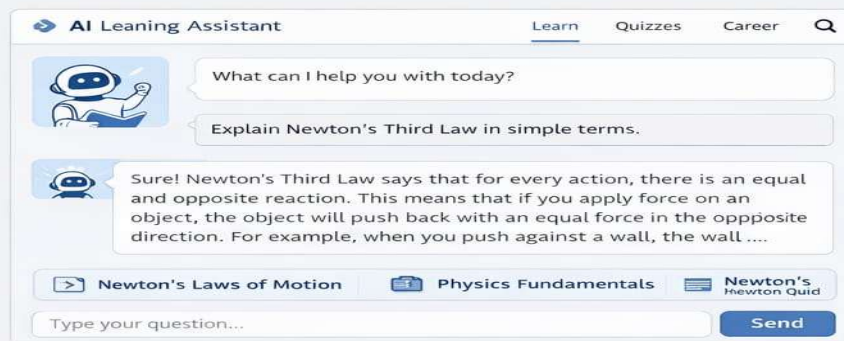
List of features offered by the solution

- AI-powered explanations of topics in simple and easy language
- Personalized learning paths based on user goals and skill level
- Automatic generation of summaries, quizzes, and practice questions
- Support for uploading notes or study material for quick revision
- Basic career guidance including skill suggestions and interview prep
- User-friendly interface that is free and accessible to everyone

Process flow diagram or Use-case diagram

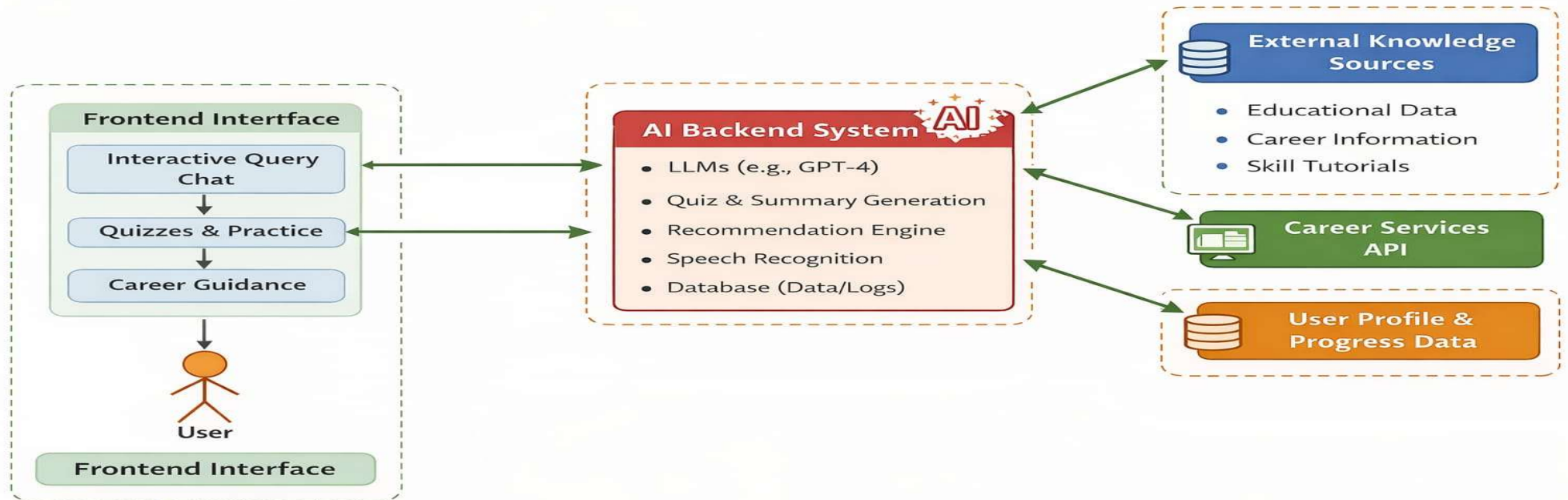


Wireframes/Mock diagrams of the proposed solution (optional)



Architecture diagram of the proposed solution

Architecture of the Proposed Solution



Technologies to be used in the solution

- **Programming Language:** Python
- **Frontend:** Streamlit
- **Backend:** FastAPI
- **Generative AI Models:** Open-source LLMs (Hugging Face)
- **AI Orchestration:** LangChain
- **Data Processing:** PDF/Text processing libraries
- **Deployment:** Hugging Face Spaces / Streamlit Cloud
- **Version Control:** Git & GitHub

Usage of AMD Products/Solutions

The proposed solution can effectively leverage AMD's high-performance computing ecosystem to support scalable and efficient Generative AI workloads. AMD Ryzen and EPYC CPUs provide strong multi-core performance, enabling fast backend processing, API handling, data preprocessing, and orchestration of AI workflows. This ensures smooth handling of multiple users and real-time interactions without performance bottlenecks.

For AI acceleration, AMD GPUs can be utilized to run and optimize Generative AI models using the ROCm (Radeon Open Compute) platform. ROCm enables efficient execution of open-source large language models, improving inference speed while maintaining cost and energy efficiency. This makes the solution suitable for both cloud deployment and local edge systems.

By using AMD's open and developer-friendly hardware and software stack, the platform benefits from better performance-per-watt, scalability, and lower operational costs. This aligns well with the project's goal of providing a free, accessible, and sustainable AI-powered learning and career assistant for everyone, while demonstrating real-world application of AMD technologies in Generative AI solutions.

Estimated implementation cost (optional)

The proposed solution is designed to be highly cost-effective, with most of the development and deployment carried out using free and open-source technologies. The application uses open-source Generative AI models, Python-based frameworks, and free hosting platforms such as Hugging Face Spaces or Streamlit Community Cloud, which eliminates licensing and infrastructure costs.

Development and testing can be performed on a standard personal laptop using free development tools. No paid APIs or proprietary software are required for the initial prototype. Any minor expenses would be limited to internet usage or optional testing resources.

Overall, the estimated cost for implementing and deploying the solution is very low, making it suitable for students and hackathon environments while remaining scalable for future enhancements.

Prototype Assets (Optional)

- GitHub Public Repository Link
- Demo Video Link (Max: 3 Minutes)

GitHub Public Repository Link

<https://github.com/JakkulaVeerababu/ADOBE-HACKATHON-GENERATIVE-AI-FOR-EVERYONE->

Add as per the requirements of the contest

Estimated Implementation Cost (Detailed Overview)

The proposed Generative AI-based Learning and Career Assistant is intentionally designed to be **highly cost-efficient**, ensuring that it can be built, deployed, and scaled without significant financial investment. The focus of the solution is on leveraging **open-source technologies, free development tools, and free-tier cloud platforms**, making it accessible for students, educators, and early-stage innovators.

All core components of the system—including AI models, backend services, frontend interface, and deployment infrastructure—are developed using freely available resources. This approach eliminates the need for expensive licenses, proprietary software, or paid APIs during the prototype and initial deployment phases.

Cost Breakdown by Components

1. Development and Tools

Programming languages such as **Python** and frameworks like **Streamlit** and **FastAPI** are completely open-source.

AI orchestration libraries (e.g., LangChain) and text/PDF processing tools are also free.

Version control and collaboration are handled using **GitHub**, which offers free repositories.

Estimated Cost: ₹0

2. Generative AI Models

- The solution uses **open-source large language models** available through platforms such as Hugging Face.
- These models can be accessed and deployed without subscription fees.
- No dependency on paid commercial APIs is required for the prototype.

Estimated Cost: ₹0

3. Infrastructure and Deployment

The application can be hosted on **Hugging Face Spaces** or **Streamlit Community Cloud**, both of which provide free hosting suitable for hackathon demos and early users.

Backend services run on free-tier compute resources.

No dedicated servers or cloud contracts are needed.

Estimated Cost: ₹0

Operational and Maintenance Costs

During the hackathon and early usage stages, operational costs remain extremely low. The system does not require continuous human monitoring or paid services. Any minimal expenses are limited to:

Internet connectivity

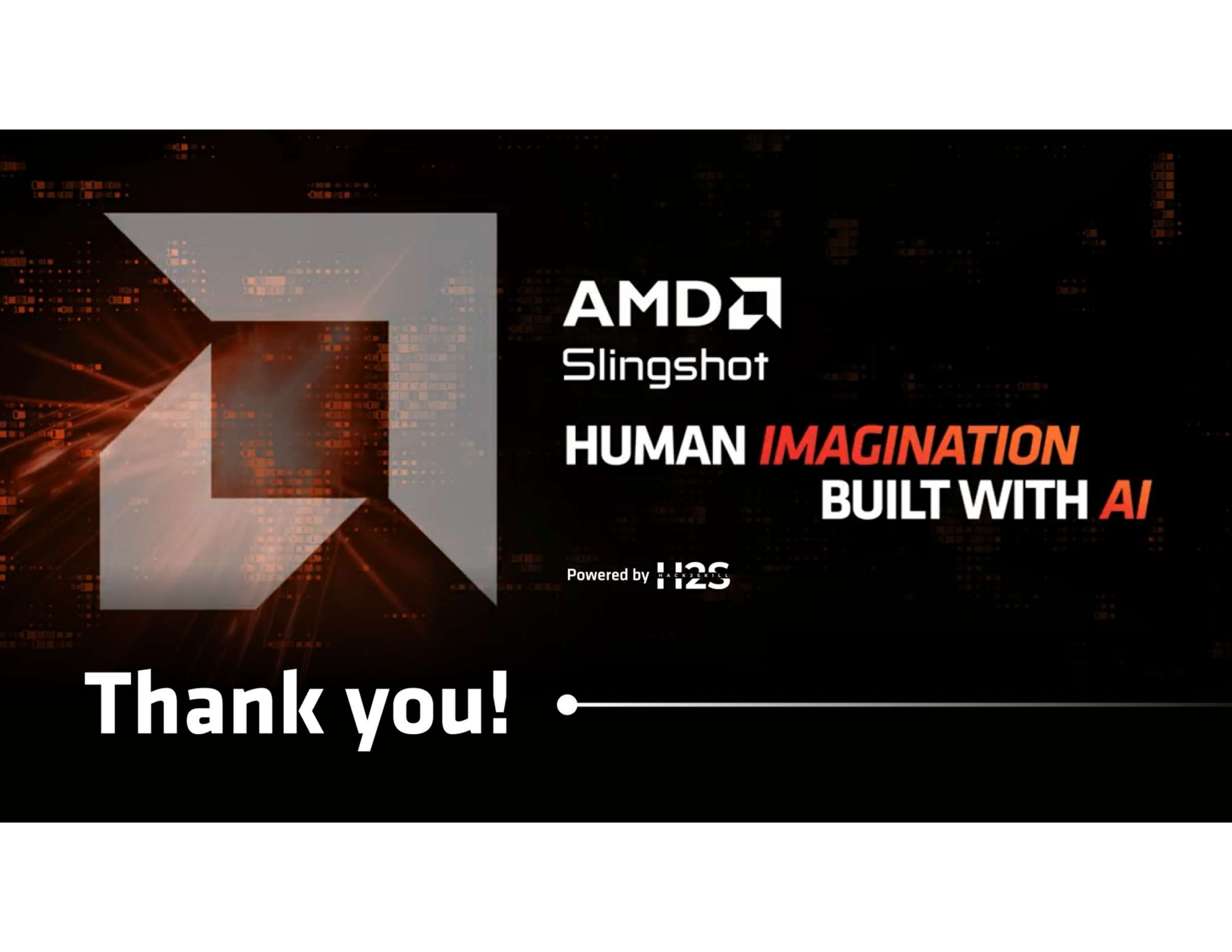
Optional extended testing or experimentation

These costs are typically already covered by personal or institutional resources.

Estimated Cost: ₹0–₹3,000 (optional)

Why This Cost Model Fits the Contest Requirements

- Encourages **innovation without financial barriers**
- Suitable for **students and first-time developers**
- Promotes **scalability and sustainability**
- Aligns with the contest vision of **inclusive and accessible AI solutions**
- Demonstrates real-world feasibility beyond the hackathon



AMD 
Slingshot

HUMAN *IMAGINATION*
BUILT WITH *AI*

Powered by  H2S

Thank you! ●—————